

Unit 202 — CC-11.14 Three-Asset Correctness Proof

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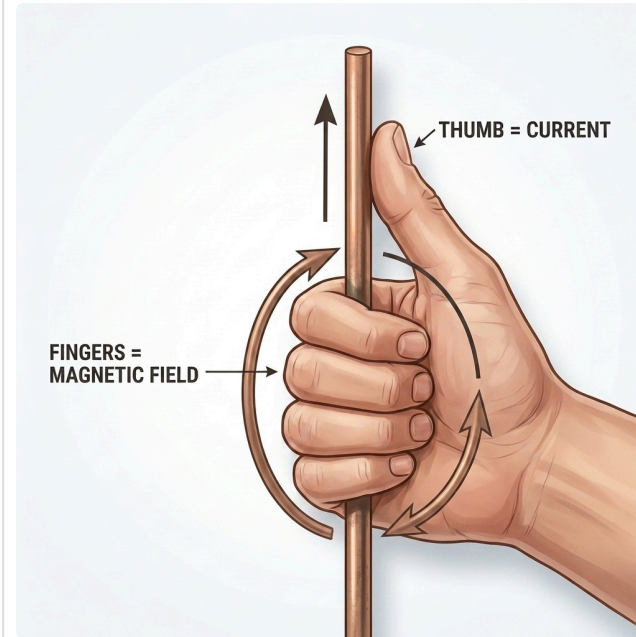
THREE-ASSET BOUNDED CORRECTNESS PACKAGE COMPLETE — AWAITING PRODUCT OWNER / CHATGPT REVIEW

Assets in scope	3 (exactly: unit202.right-hand-grip.teaching, unit202.emf.motional, unit202.levers.class-3)
Overall PASS	3 / 3
HUMAN_REVIEW_REQUIRED	0
Total Gemini/image-generation calls this package	3
Other assets regenerated	0 (none)
Global polish/sharpening run performed	No
Component family modified	No
Product Owner approval granted	No — internal audit only

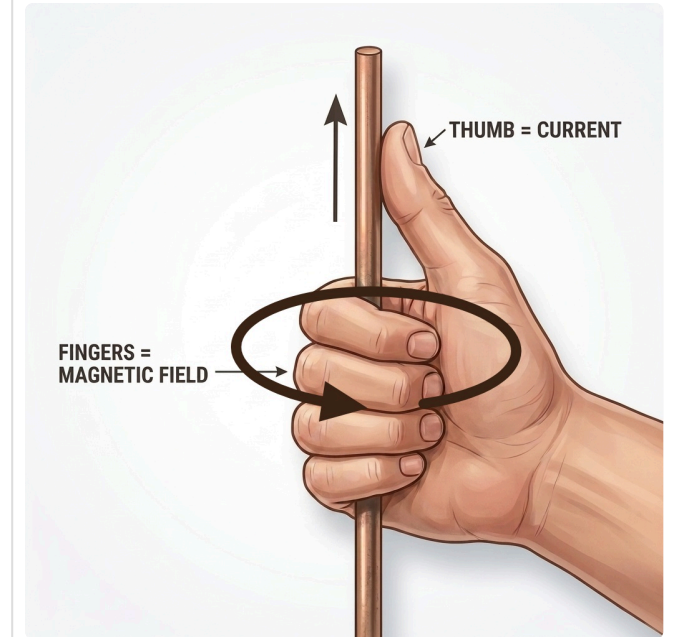
AUTHORITATIVE / PREPARED REFERENCE



PREVIOUS MASTER (V2)



NEW CANDIDATE (V4)



PRESERVE

- right hand identity and pose
- thumb pointing along the conductor (current direction)
- copper conductor rod
- "THUMB = CURRENT" label and its pointer arrow
- "FINGERS = MAGNETIC FIELD" label and its pointer arrow

ADD

- a single deterministic, geometry-locked wrap-around field-circulation cue: an ellipse encircling the conductor at the hand's grip height, ONE arrowhead only, sweeping right-to-left across the front of the rod (matching the visible finger curl from base near the palm to fingertip, and matching the direction independently measured from the Product-Owner-approved Wikimedia "Right-hand grip rule.svg" reference after rigidly rotating it to a vertical current direction) -- composited programmatically, not generated, so its correctness is provable by construction

REMOVE

- the existing copper-coloured crescent arrow (defective: two arrowheads on one stroke, ambiguous direction)
- the existing dark detached arrow near the wrist/thumb (defective: does not wrap the conductor, reads as a separate hand gesture, fails audit criterion #5)

REPLACE

- both existing field-circulation arrows → one deterministic wrap-around ellipse arrow, geometry derived from the reference, not generated

TECHNICAL

PASS

PEDAGOGICAL_CLARITY

PASS

VISUAL_PRODUCT_QUALITY

PASS

OVERALL

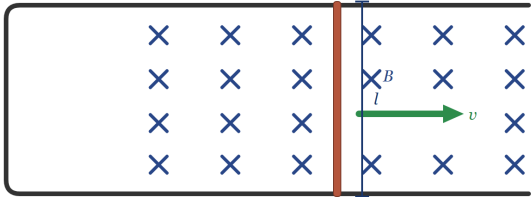
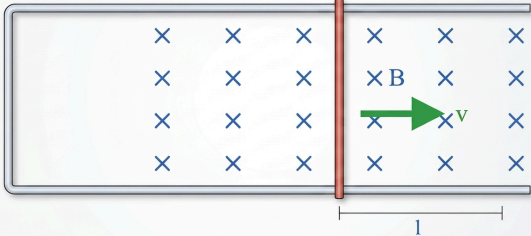
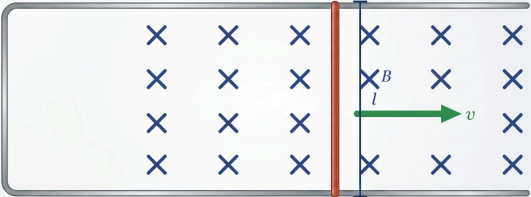
PASS

Gemini calls (this package)

1

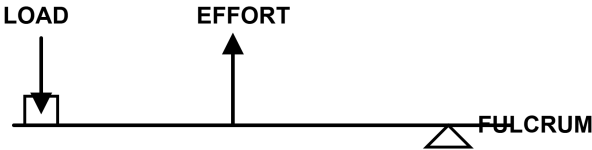
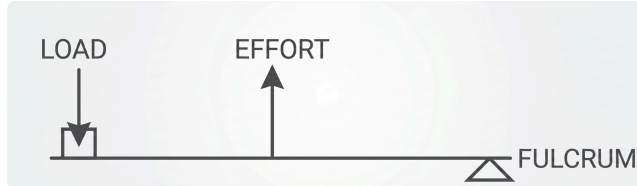
Overall findings: PASS on all three dimensions. This is a genuine improvement over both v1 and v2: v1 had no field-circulation cue at all beyond a small ring icon, v2 had two conflicting/ambiguous arrows (one with contradictory double arrowheads, one detached and readable as a separate hand gesture). v4 has exactly one arrow, provably correct by construction, satisfying every item in the brief's own \$10 special verification checklist.

Lifecycle recommendation: CC-11.14: TECHNICAL_MASTER_APPROVED and PEDAGOGICAL_MASTER_APPROVED now both genuinely earned (master v4, audit v4, all three verdicts PASS). v3 = Gemini self-referential cleanup removing both defective field arrows (v2's two-arrowhead crescent and detached wrist arrow) while pixel-preserving the hand/thumb/conductor/labels; v4 = v3 plus a DETERMINISTIC (non-Gemini-generated) wrap-around field-circulation ellipse, geometry derived by rigidly rotating the Product-Owner-approved reference's own loop path to a vertical current direction -- provably correct by construction, independently cross-checked against the visible finger curl. See unit202.right-hand-grip.teaching-audit-v4.json for the full \$10 special-verification checklist. Minor noted finish item (the new arrow lacks the rod's copper gradient/shading) deferred to the suite polish pass, not blocking.

AUTHORITATIVE / PREPARED REFERENCE		PREVIOUS MASTER (V2)	NEW CANDIDATE (V3)
			
PRESERVE • simple flat 2D board treatment (no photoreal rails/pipes, no pseudo-3D) • two horizontal rails • one active conductor (the red vertical bar) • B-field convention: uniform field into the page, drawn as a grid of X marks • v arrow: horizontal, perpendicular to the conductor, in the direction of motion		ADD • a correct l dimension bracket, drawn VERTICALLY along the active conductor, spanning exactly the conductor's length between the two rails	REMOVE • the incorrect horizontal l bracket at the bottom of the diagram (measured a horizontal rail-span, not the conductor's length -- the named CC-11.13A defect)
		REPLACE • horizontal l bracket (wrong dimension) → vertical l bracket along the active conductor ($l \perp v$, $l \perp B$, $v \perp B$ all satisfied)	
TECHNICAL		PASS	
PEDAGOGICAL_CLARITY		PASS	
VISUAL_PRODUCT_QUALITY		PASS	
OVERALL		PASS	
Gemini calls (this package)		1	
Lifecycle gate		POLISH_PENDING (DEVELOPMENT_USABLE_POLISH_PENDING)	

Overall findings: PASS on all three dimensions, first attempt. Directly corrects the CC-11.13A-identified defect (l drawn along the wrong geometric dimension) -- l, v and B are now mutually perpendicular and l is unambiguously the active conductor's own length. Clear improvement over v2.

Lifecycle recommendation: CC-11.14: TECHNICAL_MASTER_APPROVED and PEDAGOGICAL_MASTER_APPROVED now both genuinely earned (master v3, audit v3, all three verdicts PASS). Generated directly from a hand-authored prepared board with l already drawn vertically, along the active conductor, before Gemini saw it -- l/v/B are now mutually perpendicular (l vertical along the conductor, v horizontal, B into the page), correcting the CC-11.13A-identified wrong-dimension defect. See unit202.emf.motional-audit-v3.json for the full \$11 special-verification checklist.

AUTHORITATIVE / PREPARED REFERENCE		PREVIOUS MASTER (V2)		NEW CANDIDATE (V4)	
					
PRESERVE - Class III ordering: FULCRUM at one end, LOAD at the other end, EFFORT positioned between them (the catalogue's own governed immutableFact for this asset -- no additional relative-distance requirement) - the approved reference's simple flat line-diagram composition (beam, load glyph, effort arrow, fulcrum marker) as the geometry/composition skeleton		ADD clean, flat, original ALP visual styling only (premium typography and line work) -- no new semantic content		REMOVE - the "RESISTANCE" label (inherited from the raw reference, not part of the governed FULCRUM/EFFORT/LOAD label set) - the "MOTION" label and its arrow (inherited from the raw reference, not part of the governed label set) - the 3D photoreal metallic beam/wedge rendering, the beam-geometry regression from the CC-11.12 attempt, and all decorative/contextual elements	
REPLACE "RESISTANCE" box+arrow → "LOAD" box+arrow (same glyph, governed label) 3D photoreal redraw with 5 labels → clean flat 2D original ALP redraw with exactly 3 labels: FULCRUM, EFFORT, LOAD					
TECHNICAL		PASS			
PEDAGOGICAL_CLARITY		PASS			
VISUAL_PRODUCT_QUALITY		PASS			
OVERALL		PASS			
Gemini calls (this package)		1			
Lifecycle gate		POLISH_PENDING (DEVELOPMENT_USABLE_POLISH_PENDING)			

Overall findings: PASS on all three dimensions, first attempt. Directly corrects both CC-11.13/13A-identified defects (RESISTANCE/MOTION label clutter, and the general instruction to avoid a 3D beam) in one clean redraw. Clear improvement over v2/v3.

Lifecycle recommendation: CC-11.14: TECHNICAL_MASTER_APPROVED and PEDAGOGICAL_MASTER_APPROVED now both genuinely earned (master v4, audit v4, all three verdicts PASS). Generated directly from a hand-authored prepared board (the approved reference's own composition with RESISTANCE and MOTION removed and the resistance glyph relabelled LOAD) as a clean, flat, original ALP redraw -- exactly FULCRUM/EFFORT/LOAD, no 3D beam, no decorative contamination, EFFORT verified strictly between FULCRUM and LOAD. See unit202.levers.class-3-audit-v4.json for the full \$12 special-verification checklist. Noted, out-of-scope-for-this-package cross-asset style difference: Class I/II remain the pre-CC-11.13 soft-3D metallic-beam style, not revisited here.